

# How to Fake a Badge like a Pro: 11 Tips to Counterfeiting Event Credentials

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Who am I?

Russell Phillips

**SXSW** - Premiere Cultural Festival

20+ Years in Information Security & Event Access Control

Frontline of anti-counterfeiting & anti-fraud at the world's  
premiere cultural festival

## Why this talk?

- Too much security by obscurity
- Too many bad counterfeits
- Concepts are broadly applicable to information security systems

## Why are Event Credentials an Interesting Challenge?

- Fast & Chaotic
- Secretive, security by obscurity
- Broadly applicable principles of security design/defeat
- High value & high challenge

## Credential Anti-counterfeiting Tech

- Brief overview
- See talk notes for further info

# Material

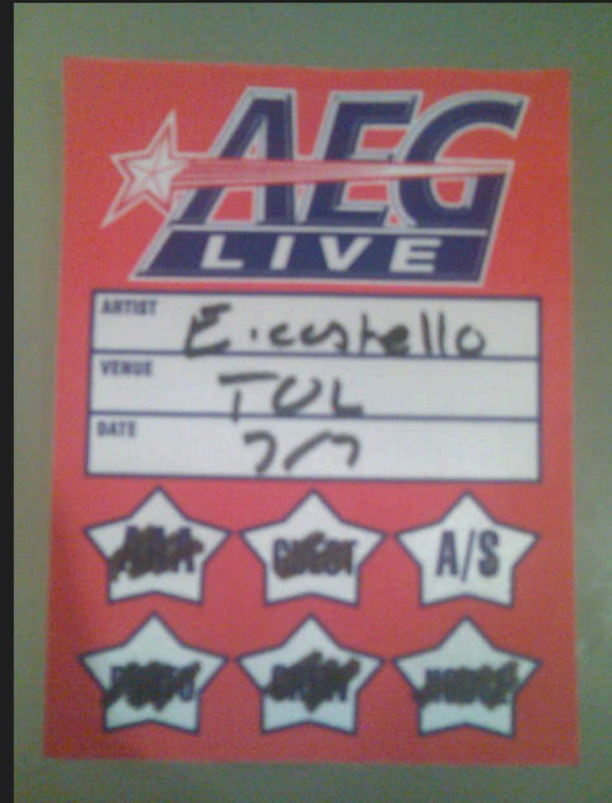


Credit: goblinbox\_(queen\_of\_ad-hoc\_bento) [CC BY 2.0.]

# Visual Elements



Credit: Kmeron [CC BY-NC-ND 2.0]



Credit: Mark Hillary [CC BY 2.0]



## Other Optical Elements



Credit: Daniel Clayton Greer [CC BY-NC-ND 2.0]



Credit: (T)imotheop [CC BY-NC-ND 2.0]



# Multi-element Composites



Credit: SqueakyMarmot [CC BY-SA 2.0]



Credit: cnadia [CC BY-NC-SA 2.0]

# Referencial



Credit: Jerry "Woody" [CC BY-SA 2.0]



# Cryptography



Credit: Mayo Nissen [CC BY-NC-SA 2.0]



## 11 Tips for Counterfeiting Event Credentials

- What do you need to get into?
- Are there better ways in?
- How are credentials checked?
- Countersurveillance

## Tip #1 - Look the Part

- Play dress up
- Bring props
- Match attitude & demeanor

## Tip #2 - Retouch & Recreate

- Photocopies stand out
- Matched level of detail is better than element accuracy
- Clear & clean prints



## Tip #3 - Touch Test

- Fingers are nearly as important as eyes
- Determine when credentials might be handled and how

## Tip #4 - Stay in Your Lane

- You dressed for a part, don't act out of it.
- Becoming someone's problem is the quick way to getting in worse trouble.

## Tip #5 - Accessorize!

- Ensure you have everything associated with a valid credential
- Inventory items that accompany properly issued credentials
- Receipts, stickers, etc

## Tip #6 - Colors Matter

- Color errors stand out more than detail mistakes
- Use inconspicuous items like clothing or pamphlets as color references in photos

## Tip #7 - Consider Material (Laminates aren't pouches)

- Pouches versus laminates
- Thermal paper versus printer paper
- Staple, clasp, etc color



## Tip #8 - Consider Process (Cutting isn't tearing)

- Cut versus torn
- Stamped versus printed
- Surface printed versus dyed cloth
- Relief versus intaglio

## Tip #9 - Match photo quality

- Is there a consistent background?
- Photos often printed separately
- Downgrade quality if needed

## Tip #10 - Scope Serial Numbers

- Research required
- Depending on your plan, passing but invalid might be ok
- Sometimes invalid might be better than already used

## Tip #11 - Age & Condition

- Day 1 versus Day 11
- Multi-event credentials
- Credentials issued ahead of the event

## Case Studies

- 5 examples of credentials through history
- Review elements
- Call out some (but not all) of the tips that apply

## Case Studies - Tips for all cases

#1 - Look the part

#3 - Touch test

#4 - Stay in your lane

#5 - accessorize!





## Case #1 - WW2 Era US Navy ID Card

### Tip #7 - Consider Material



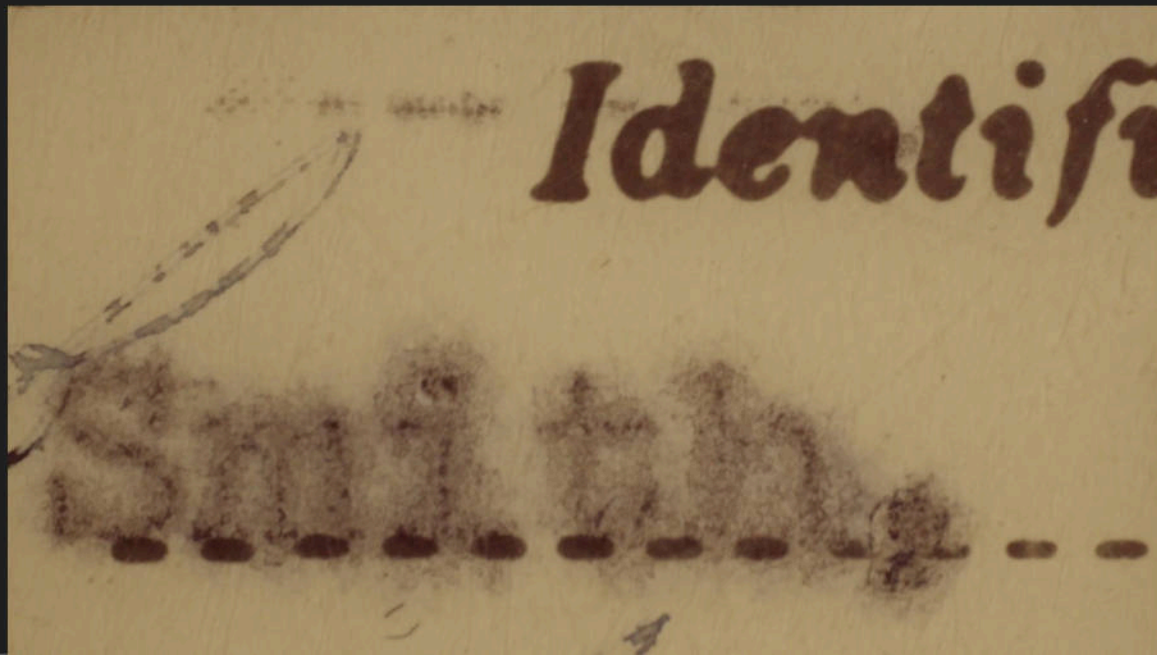
## Case #1 - WW2 Era US Navy ID Card

### Tip #9 - Match photo quality



## Case #1 - WW2 Era US Navy ID Card

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Tip #11 - Age & Condition

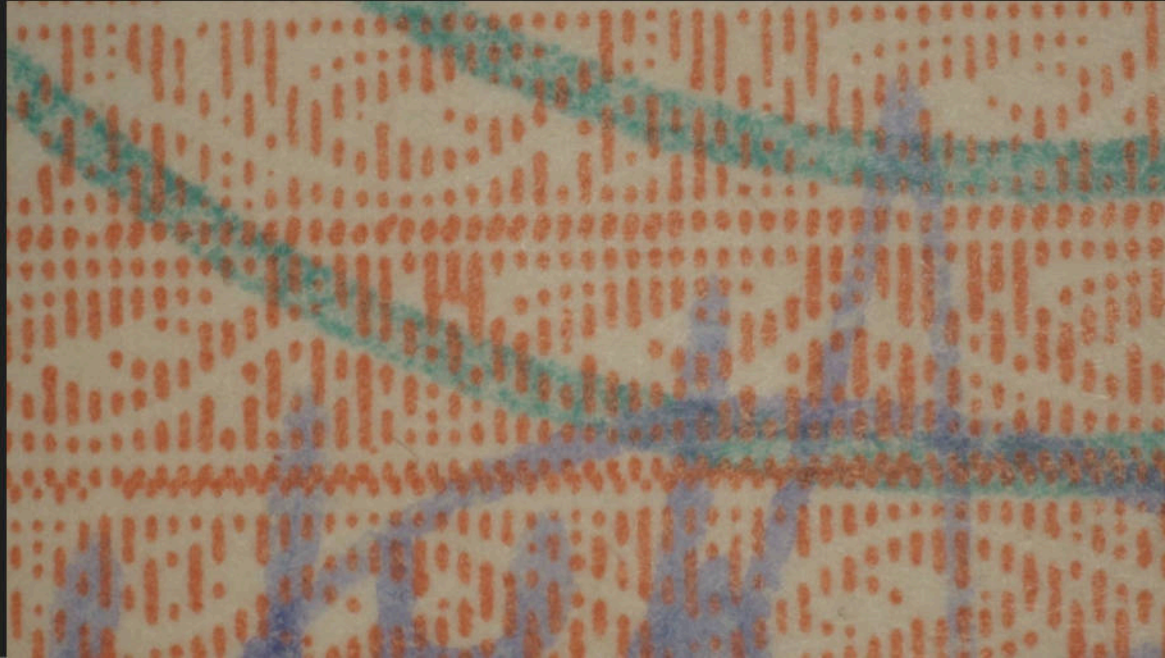
Tip #7 - Consider Material

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## Case #2 - 1966 NATO Delegation Pass

### Tip #6 - Colors Matter





## Case #2 - 1966 NATO Delegation Pass

### Tip #11 - Age & Condition



Case #2 - 1966 NATO Delegation Pass

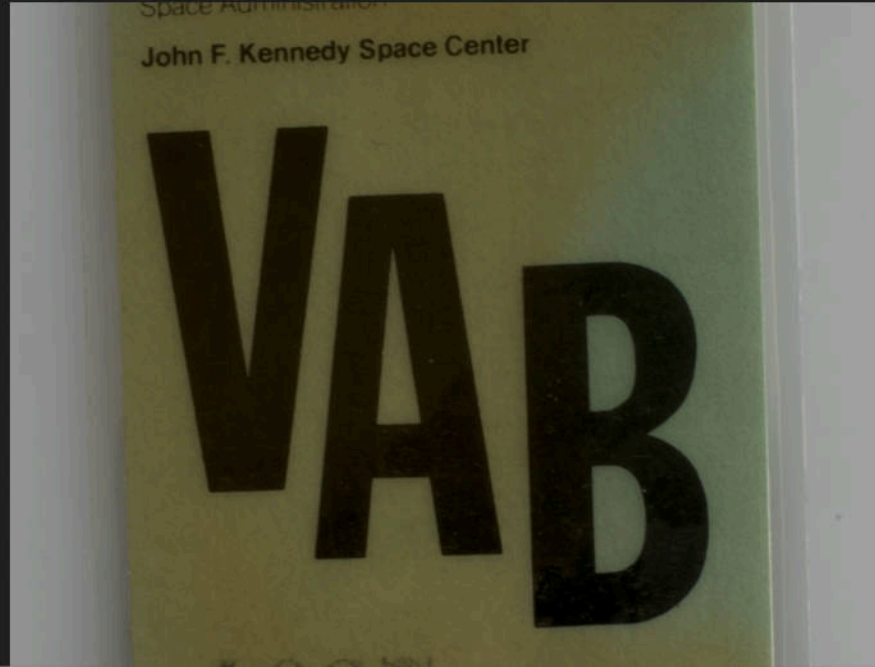
Tip #6 - Colors Matter

Tip #11 - Age & Condition



## Case #3 - NASA Vehicle Assembly Building Pass

### Tip #6 - Colors Matter



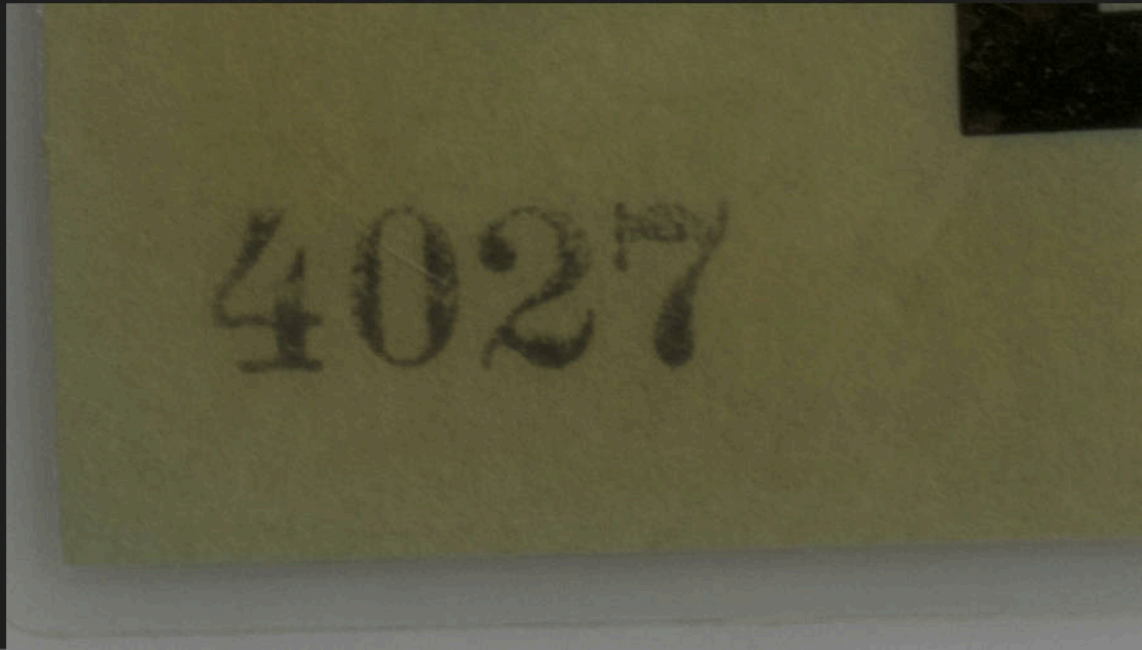
## Case #3 - NASA Vehicle Assembly Building Pass

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## Case #3 - NASA Vehicle Assembly Building Pass

### Tip #10 - Scope Serial Numbers



Tip #6 - Colors Matter

Tip #7 - Consider Material

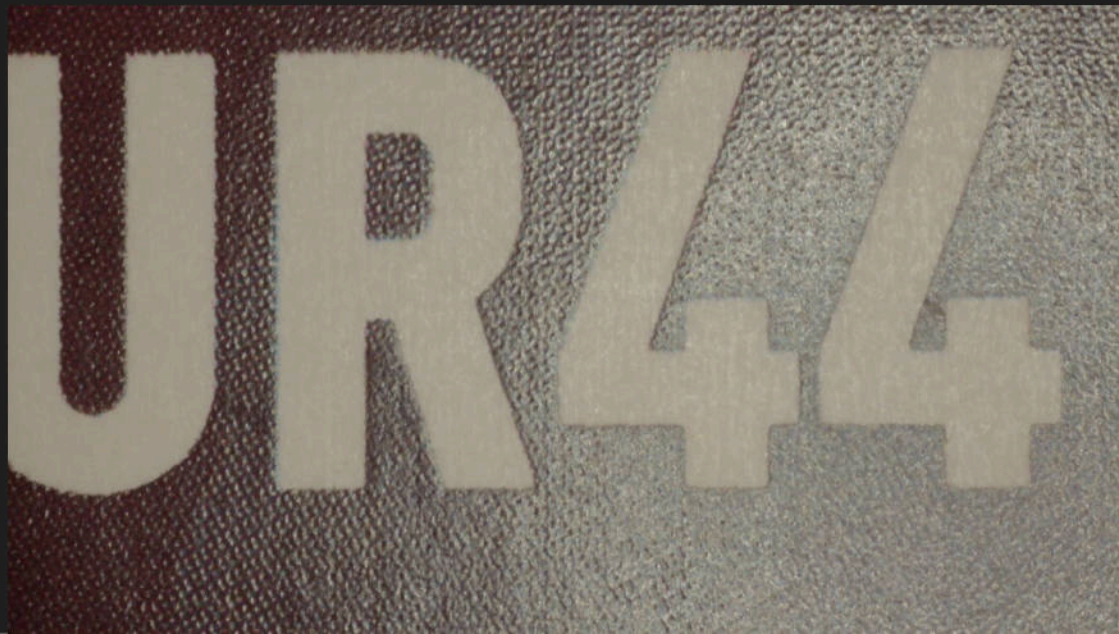
Tip #10 - Scope Serial Numbers





## Case #4 - 2021 NFL Game Ticket

### Tip #2 - Retouch & Recreate



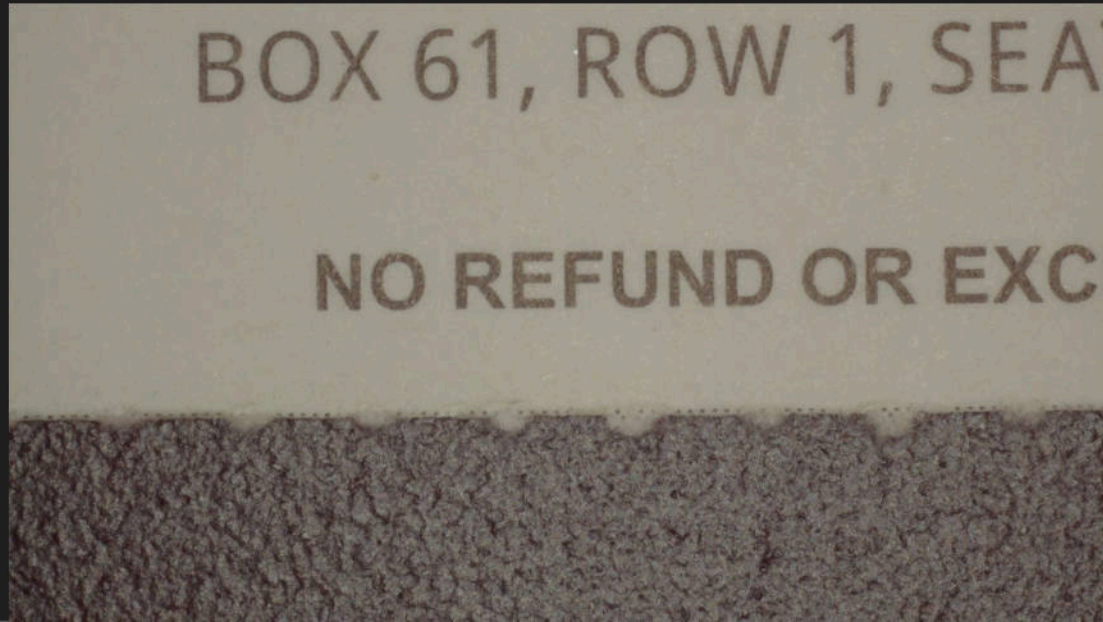
## Case #4 - 2021 NFL Game Ticket

### Tip #6 - Colors Matter



## Case #4 - 2021 NFL Game Ticket

### Tip #8 - Consider Process



## Case #4 - 2021 NFL Game Ticket

Tip #2 - Retouch & Recreate

Tip #6 - Colors Matter

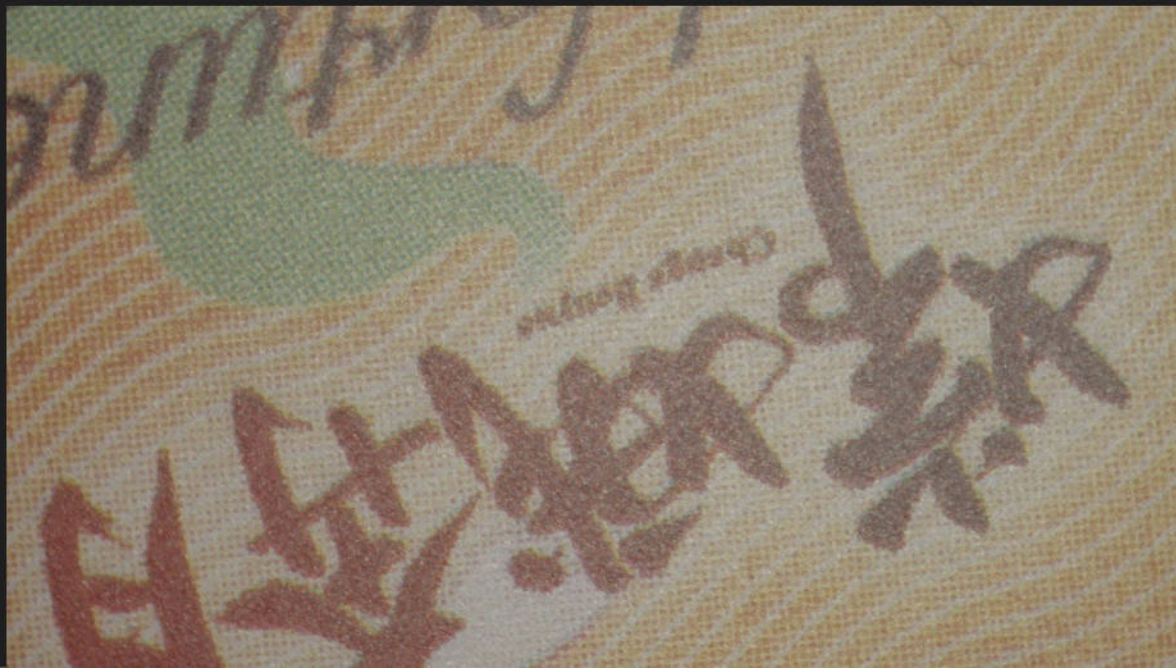
Tip #8 - Consider Process





## Case #5 - 2023 Mid-Autumn Festival Ticket

### Tip #2 - Retouch & Recreate





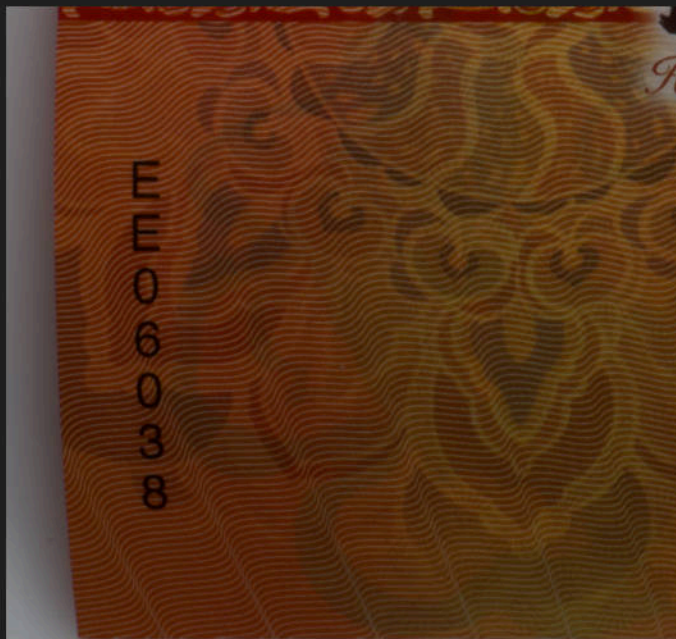
Case #5 - 2023 Mid-Autumn Festival Ticket

Tip #6 - Colors Matter



## Case #5 - 2023 Mid-Autumn Festival Ticket

### Tip #10 - Scope Serial Numbers



Tip #2 - Retouch & Recreate

Tip #6 - Colors Matter

Tip #10 - Scope Serial Numbers

## Case Study Review

- Real events have more challenges
- Use multiple credentials as needed
- If they never check your credential, they can't spot a fake

## Final Thoughts

- Think about people: users & checkers
- Think about process: physical & rules
- Think about systems: human, physical, social (and digital)

Thanks!

Questions or comments? Send me an email: [connect@russell.computer](mailto:connect@russell.computer)